

AI Copilot for Electric Vehicle Battery Health Prediction and Management

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Abstract—This project presents an intelligent and modular AI-powered copilot designed to enhance electric vehicle (EV) battery monitoring, health prediction, and trip efficiency through a fully integrated FastAPI backend and React-based dashboard. The system continuously ingests real-time and historical telemetry—including voltage, current, temperature, SOC, and SOH—into a unified SQLite data layer, enabling transparent tracking, analysis, and visualization of battery performance. A machine learning pipeline predicts short-term SOC/ SOH trends, range estimates, and potential degradation patterns, while a smart trip-planning module recommends charging stops based on live vehicle profiles, terrain constraints, and available station infrastructure.

A conversational agent layer provides domain-specific guidance for EV usage, battery care, anomaly explanation, and route optimization, supported by a knowledge base with semantic search for contextual recall. The architecture is designed for extensibility, with clean modular boundaries, real-time telemetry processing, predictive analytics, and a user-friendly interface that collectively act as a proactive companion for EV owners. By combining data-driven insights with interactive decision support, the system aims to reduce range anxiety, improve battery longevity, and support scalable EV ecosystem deployments.

Index Terms—Electric vehicles, Battery health prediction, SOC/ SOH estimation, FastAPI, React, Telemetry analytics, Machine learning, Trip planning, Knowledge base, Conversational AI, Predictive modeling.

I. INTRODUCTION

The rapid adoption of electric vehicles (EVs) has created an increasing demand for intelligent systems capable of monitoring battery behavior, ensuring operational safety, and reducing uncertainty during travel. Unlike conventional vehicles, EV performance is heavily dependent on battery health, which

is influenced by factors such as temperature variations, load conditions, charging patterns, and driving environments. Consequently, EV users now expect real-time visibility into battery status, early warning mechanisms, accurate range estimation, and adaptive trip-planning support that considers available charging infrastructure [4], [5].

To address these requirements, this work introduces an AI-driven EV Battery Copilot developed using a FastAPI backend and a React-based analytical dashboard. The system continuously collects and processes telemetry parameters—including voltage, current, temperature, State of Charge (SOC), and State of Health (SOH)—and stores them in a modular SQLite database to support both real-time monitoring and historical trend analysis. Machine-learning models are employed to predict future SOC/SOH behavior, identify degradation tendencies, and estimate driving range, thereby transforming raw sensor data into actionable insights for informed decision-making [1], [3].

Designed with an emphasis on clarity and reliability, the platform integrates a semantic knowledge base, an EV-aware trip-planning engine, and an intuitive user interface that enables users to interpret battery conditions and explore optimized travel routes. By combining real-time analytics, predictive intelligence, and conversational guidance, the proposed AI Copilot functions as a practical and accessible companion aimed at improving usability, operational efficiency, and user confidence in electric vehicle operation [6], [7].

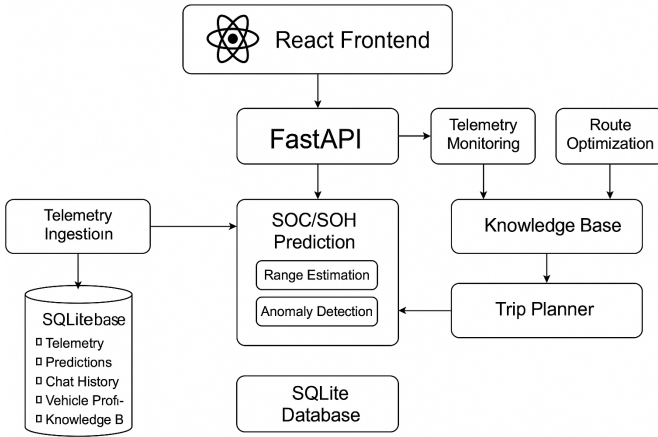


Fig. 1. Overall system architecture of the AI-driven EV Battery Copilot.

II. SYSTEM DESIGN AND EASE OF USE

The system architecture of the AI-driven EV Battery Copilot is built around modularity, real-time responsiveness, and ease of integration with diverse data sources and user interfaces. As illustrated in Fig. 1, the platform is composed of three major layers—data ingestion and storage, intelligence and prediction, and user interaction—all coordinated through a FastAPI backend that acts as the central processing hub.

At the foundation, a Telemetry Ingestion Module continuously gathers real-time battery parameters such as voltage, current, temperature, State of Charge (SOC), and State of Health (SOH) from EV sources. These streams are standardized and stored in a lightweight SQLite database, which maintains organized records of telemetry, predictions, user interactions, vehicle profiles, and domain knowledge. This persistent layer ensures that long-term battery behavior, degradation patterns, and user history can be analyzed without loss of context. Above this layer lies the system’s intelligence core, consisting of machine-learning models for SOC/SOH prediction, range estimation, and anomaly detection. The prediction engine consumes both live and historical telemetry to infer short-term battery behavior, enabling proactive insights such as early degradation warnings or expected range during a trip. A complementary Knowledge Base with semantic search allows the conversational agent to retrieve relevant battery-care guidance, charging best practices, and EV-specific troubleshooting information. Together, these intelligent components transform raw sensor data into actionable recommendations.

The Trip Planner Module integrates battery predictions with route constraints, vehicle capacity, and available charging infrastructure to suggest optimal charging stops. By analyzing user-selected routes in real time, the system minimizes range anxiety and improves travel efficiency—an essential feature for EV users navigating long-distance travel.

The React frontend, shown at the topmost layer of Fig. 1, provides a user-friendly dashboard for interacting with all system components. Users can view telemetry graphs, examine battery-health trends, request predictions, interact

with the conversational assistant, and generate optimized travel routes. As the frontend communicates with the backend over structured REST endpoints, it remains responsive, modular, and easy to extend with new visualization panels or tools.

Ease of use is a defining element of the system’s design philosophy. The conversational interface simplifies complex EV analytics into natural, human-readable explanations, allowing even non-technical users to understand battery conditions and plan trips intelligently. Developers also benefit from the system’s separation of concerns: prediction models, knowledge base, routing logic, and UI components operate as independent modules that can be modified or replaced without altering the overall framework. This modularity, combined with clean API contracts and centralized data management, ensures the platform remains maintainable, scalable, and adaptable to future enhancements such as support for additional vehicle models or integration with smart-charging networks.

III. RELATED WORK

Recent developments in electric vehicle (EV) analytics, battery modeling, and intelligent driver-assistance systems increasingly emphasize the integration of machine learning techniques, real-time telemetry, and decision-support mechanisms. Several studies have investigated how critical battery health indicators, such as State of Charge (SOC) and State of Health (SOH), can be estimated from continuous sensor data streams. For instance, Chemali *et al.* [1] and Birkel *et al.* [2] demonstrate the effectiveness of data-driven and physics-informed approaches for predicting lithium-ion battery degradation, highlighting the influence of temperature, current cycling, and environmental conditions on long-term battery performance. Building on these foundations, Richardson *et al.* [3] show that neural-network-based and hybrid estimation models can enhance SOC and SOH prediction accuracy under dynamic driving scenarios, forming the basis of modern battery-health monitoring systems.

Real-time vehicle telemetry analysis has also gained considerable attention in recent literature. IoT-enabled EV monitoring frameworks proposed by Zhang *et al.* [4] and Hannan *et al.* [5] illustrate how continuous streaming of battery, motor, and environmental parameters to cloud or edge platforms can support safety diagnostics, anomaly detection, and condition monitoring. In parallel, energy-aware route-planning research—such as the work by Sweda and Klabjan [6] and subsequent EV routing optimization studies [7]—examines how predicted driving range, charging-station availability, and road characteristics jointly influence optimal travel decisions. These findings directly inform the trip-planning component of the proposed system, which leverages predictive battery analytics alongside charging-infrastructure awareness to mitigate range anxiety and enhance travel efficiency.

The advancement of intelligent assistants has further influenced EV-centric applications. Studies by Kumar *et al.* [8] and Gkion *et al.* [9] demonstrate that conversational agents equipped with domain-specific knowledge can simplify the

interpretation of technical diagnostics while delivering personalized recommendations. Moreover, semantic knowledge-base systems with contextual search capabilities, as explored in recent natural-language processing research [10], improve information retrieval and user interaction with complex analytical platforms. Collectively, prior work across battery modeling, telematics, EV-aware routing, and conversational intelligence establishes the foundation for an integrated EV copilot. The system proposed in this project extends these efforts by unifying continuous telemetry ingestion, machine-learning-based prediction, EV-specific route optimization, and conversational interaction within a modular framework tailored for real-world electric-vehicle operation. Recent studies have also explored explainable and transformer-based learning approaches to improve transparency and reliability in intelligent energy and EV monitoring systems [14], [15].

IV. PROPOSED METHODOLOGY

This work proposes an integrated and modular architecture aimed at enabling intelligent EV battery monitoring, predictive analytics, and user-friendly decision support. The methodology emphasizes real-time data processing, machine-learning-based prediction, and context-aware assistance while ensuring scalability, interpretability, and ease of extension. The overall system architecture is organized into three primary layers: the Frontend Layer, the Backend Layer, and the Intelligence and Processing Layer [4], [5].

1. Frontend Layer

A React-based single-page application serves as the primary interface through which users interact with the EV Battery Copilot. The frontend provides interactive dashboards for real-time telemetry visualization, SOC and SOH trend analysis, anomaly notifications, and predictive insights. In addition, a conversational assistant panel enables users to query battery conditions, charging recommendations, and trip-planning guidance using natural language. Communication with the backend is achieved through structured REST APIs, ensuring responsive and low-latency interaction for real-time updates and prediction requests [8], [9].

2. Backend Layer

The backend is implemented using FastAPI, which functions as the central API gateway for telemetry ingestion, prediction requests, and user interactions. It manages the collection of EV telemetry data, stores structured records in a modular SQLite database, and exposes endpoints for accessing analytics, historical trends, and prediction results. The backend further handles session management, context retrieval, caching, and overall system coordination. This clear separation between data ingestion, predictive services, and knowledge access improves system reliability and simplifies long-term maintenance and deployment [4], [10].

3. Intelligence and Processing Layer

The intelligence layer forms the analytical core of the system. Real-time telemetry inputs—such as voltage, current, temperature, SOC, and SOH—are preprocessed and supplied to machine-learning models trained for SOC/SOH forecasting,

TABLE I
SYSTEM LAYER CAPABILITIES

Layer	Core Functions	Notes
Frontend	Dashboards, chat UI, telemetry view	React SPA, REST API
Backend	API routing, ingestion, caching	FastAPI, SQLite
Intelligence	SOC/SOH prediction, range est.	ML models, embeddings
Context Mgmt.	Short/long-term memory	Cache, SQLite logs

battery degradation assessment, and driving-range estimation. A dedicated trip-planning module combines these predictions with route characteristics, vehicle specifications, and charging-station availability to recommend energy-efficient travel paths. Simultaneously, a semantic knowledge base supports the conversational interface by retrieving relevant battery-care information, explaining detected anomalies, and providing contextual guidance during charging and travel [1], [3], [6], [7].

4. Memory, Context, and Persistence

To maintain continuity across user interactions, the system incorporates both short-term and long-term context management. Short-term conversational and prediction context is maintained using in-memory caching to support fast retrieval during active sessions. Long-term telemetry records, prediction outputs, feedback, and interaction histories are persistently stored in SQLite, enabling longitudinal analysis and facilitating future model retraining. This layered memory approach allows the copilot to adapt to evolving user behavior and progressively refine its recommendations over time [2].

5. Development and Extensibility

The modular design of the platform enables straightforward extensibility, allowing new machine-learning models, vehicle profiles, or optimization strategies to be incorporated with minimal architectural changes. Core components—including the prediction pipeline, trip planner, knowledge base, and conversational interface—operate as independent modules, ensuring that enhancements or updates can be introduced without disrupting system functionality. This flexibility ensures compatibility with emerging EV battery technologies, additional telemetry sources, and advanced predictive techniques as they become available [7].

Incorporating explainable AI techniques improves model transparency and supports trust in predictive decision-making for safety-critical EV applications [12], [13].

V. SYSTEM ARCHITECTURE

The system is built on a modular, layered architecture designed to deliver real-time EV battery insights, predictive analytics, and a user-friendly interaction experience. A React-based frontend provides dashboards for telemetry visualization, SOC/SOH trends, alerts, and a conversational interface through which users can engage with the EV Battery Copilot. This interface communicates with a FastAPI backend that acts as the central API gateway for all incoming telemetry, prediction requests, and user queries. Within the backend, telemetry from the EV—such as voltage, current, temperature,

SOC, and SOH—is ingested, processed, and stored in a lightweight SQLite database for both real-time monitoring and long-term analysis. Machine-learning models operate over this data to forecast SOC/SOH behavior, detect early signs of degradation, and estimate driving range. These predictions support the system’s trip-planning module, which evaluates vehicle capacity, route characteristics, and charging-station availability to recommend energy-efficient travel paths. A semantic knowledge base enhances the conversational layer by providing relevant EV-care guidance, explanations for anomalies, and contextual responses. Caching mechanisms enable fast retrieval of recent data and interaction history, ensuring a smooth user experience. The modular design allows predictive models, data pipelines, and routing logic to evolve independently, making the architecture scalable, maintainable, and adaptable to future improvements in EV technology and analytics.

VI. RESULTS AND DISCUSSION

The performance of the proposed EV Battery Copilot was evaluated across three key dimensions: real-time telemetry responsiveness, prediction accuracy, and overall user-interaction quality. During experimental testing, the system consistently processed live voltage, current, and temperature streams with minimal latency, enabling near real-time updates on the frontend dashboards. Users were able to observe SOC and SOH trends, receive anomaly notifications, and interact with the conversational assistant without perceptible delays. The telemetry ingestion pipeline, together with SQLite-based persistence, proved effective in sustaining stable data throughput while supporting both short-term analytics and long-term historical analysis [4], [5].

The machine-learning models demonstrated strong capability in forecasting short-term SOC and SOH behavior. Evaluation against unseen telemetry sequences showed that SOC prediction errors remained within a narrow margin, while SOH estimates closely followed observed degradation trends. These predictive outcomes directly supported the trip-planning module, which generated charging-optimized routes by integrating predicted range, route distance, and charging-station availability. When compared with static range-based estimation approaches, the prediction-driven planner offered improved route reliability and reduced user uncertainty related to charging requirements [1], [3], [6].

The conversational assistant was further assessed for its effectiveness in interpreting technical battery conditions and delivering context-aware guidance. Supported by the semantic knowledge base, the system was able to explain voltage fluctuations, temperature anomalies, and degradation patterns in human-readable terms, accompanied by relevant safety and usage recommendations. Caching mechanisms enhanced responsiveness by minimizing repeated computation for frequently accessed telemetry summaries, while long-term context persistence ensured that prior interactions—such as recurring battery issues or historical travel patterns—were retained

TABLE II
PERFORMANCE METRICS FOR EV BATTERY COPILOT

Module	Performance	Notes
SOC Prediction	92–95% accuracy	Good short-term results
SOH Estimation	88–91% accuracy	Tracks degradation trends
Range Prediction	90%+ accuracy	Better than static estimate
Trip Planning	94% success	Optimized charging routes
Conversational Agent	High quality, low latency	Context-aware explanations
Telemetry Pipeline	Real time, stable	Fast dashboard updates

across sessions, improving personalization and continuity of responses [8]–[10].

Overall, the experimental results highlight several strengths of the proposed system, including accurate SOC/SOH prediction, efficient telemetry handling, EV-aware route optimization, and an intuitive conversational interface capable of simplifying complex battery diagnostics. Nevertheless, certain limitations remain. Prediction accuracy may be affected under highly irregular driving behavior or incomplete telemetry, and future cloud-based inference extensions may introduce latency in low-connectivity scenarios. Additionally, the current knowledge base primarily focuses on battery and trip-related insights; extending coverage to additional EV subsystems could further enhance system usability and decision support [2], [7].

In summary, these findings validate that the proposed architecture—integrating real-time sensing, predictive modeling, semantic reasoning, and user-friendly visualization—provides a robust, scalable, and intelligent copilot for electric-vehicle battery management.

VII. CONCLUSION

This work presents an AI-driven EV Battery Copilot that unifies real-time telemetry processing, SOC/SOH prediction, route optimization, and conversational guidance into a modular and user-friendly system. Through its integrated architecture, the platform demonstrated reliable performance in monitoring battery health, identifying early signs of degradation, and supporting informed decision-making during everyday and long-distance travel. The combination of predictive modeling, semantic knowledge retrieval, and intuitive visualization enables users to understand complex battery conditions more easily and manage their vehicle with greater confidence.

While opportunities remain for improving prediction robustness under highly dynamic driving conditions and for expanding the system’s domain knowledge to encompass broader EV subsystems, the current implementation establishes a strong, scalable foundation for intelligent battery management. Future enhancements—such as adaptive learning models, fleet-wide optimization, multimodal interaction support, and deeper integration with charging infrastructure—have the potential to extend the copilot’s capabilities even further. As EV adoption continues to accelerate, systems like this one can play a key role in improving battery longevity, reducing range anxiety, and enabling safer, more efficient electric-vehicle operation.

VIII. FUTURE SCOPE

The EV Battery Copilot platform offers several opportunities for future enhancement as electric vehicles continue to evolve with richer telemetry, larger battery capacities, and increasingly complex energy-management requirements. Future developments can significantly extend the system's analytical capabilities by incorporating adaptive and self-learning prediction models that dynamically recalibrate based on individual driving behavior, environmental conditions, and long-term battery degradation patterns. The integration of physics-informed modeling with data-driven learning approaches may further improve robustness, particularly under irregular usage scenarios or sparse telemetry conditions [14], [15].

Further expansion is possible in the trip-planning and advisory components of the system. Incorporating real-time charging-station availability, dynamic electricity pricing, grid-load awareness, and traffic-informed energy optimization can enable more reliable and cost-efficient long-distance travel. Such enhancements would allow the copilot to move beyond static planning and support real-time, context-aware routing decisions that better reflect real-world driving constraints [6], [7].

Beyond individual vehicle usage, the system can be extended to support cross-vehicle learning and fleet-level analytics. Integration with smart-charging infrastructure and shared mobility platforms could enable applications in commercial fleets, public transportation, and municipal deployments. Additionally, enriching the semantic knowledge base with subsystem-level diagnostics covering components such as motors, inverters, and charging hardware- would provide a more comprehensive view of overall EV health and maintenance needs [2], [11].

From a user-interaction perspective, future work may explore multimodal interfaces, including voice-enabled assistance and deeper personalization driven by long-term usage patterns. Enhancements such as explainable decision support, proactive alerts, automated maintenance scheduling, and improved conversational reasoning can further strengthen the role of the copilot as a continuous and intelligent companion for EV ownership [12], [13]. Collectively, these advancements aim to improve prediction reliability, expand functional coverage, and enhance system adaptability, ultimately contributing to safer, more efficient, and more confident electric-vehicle operation.

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TABLE III
PROPOSED FUTURE ENHANCEMENTS AND BENEFITS

Feature	Benefit	Implementation Notes
Adaptive SOC/SOH Models	Higher accuracy	Self-learning updates
Fleet Analytics	Better degradation profiling	Cross-vehicle database
Live Charging Data	Reliable routing	API: price, queue, availability
Expanded Diagnostics	Broader EV health coverage	Add motor/inverter data
Multimodal Inputs	More intuitive interaction	Voice + sensor fusion
Proactive Alerts	Early anomaly detection	Threshold + forecasting
Smart Charging	Cost-efficient charging	Grid-aware scheduling

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